

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING

Kathmandu Engineering College.
Kalimati, Kathmandu.

Major Project Report On:
Follow me bot



To:

Department of Electronics and Communication Engineering

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ABSTRACT

A mobile robot is a machine controlled by software that use sensors and other technology to identify its surroundings and move around its environment. Mobile robots function using a combination of artificial intelligence (AI) and physical robotic elements, such as wheels, tracks and legs.

Human following is one of the fundamental functions in human–robot interaction for mobile robots. In the last few years, human-following robots have become a highly active research area in various mobile robotics and computer vision applications.

Our system hopes to follow a specific person whom has been detected by camera. The robot must be able to reliably track the target person within the surrounding world until person press the stop button.

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LIST OF ABBREVIATIONS

ARM	Advance RISC Machine
BGR	Blue Green Red
CPU	Central Processing Unit
CSI	Camera Serial Interface
DC	Direct Current
DOS	Disk Operating System
DSI	Display Serial Interface
GND	Ground
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communications
HDMI	High-Definition Multimedia Interface
IC	Integrated Circuit
IDE	Integrated Development Environment
IOT	Internet of Things
IR	Infrared
LAN	Local Area Network
LiPo	Lithium Polymer
LRF	Laser Range Finder
MCAD	Mechanical Computer-Aided Design
RFID	Radio Frequency Identification

RISC	Reduced Instruction Set Computer
SD	Secure Digital
SIMCOM	Simultaneous Communication
SSD	Single-Shot multibox Detection
TV	Television
USART	Universal Synchronous/Asynchronous Receiver/Transmitter
USB	Universal Serial Bus

CHAPTER 1

INTRODUCTION

1.1 Background Theory

As of the 21st century, from automated industrial lines to simple daily use of electronic devices, the mobile robot has become a hobby for those robot fanatics. The basic functions of a mobile robot include the ability to move and explore, transport payloads, or revenue producing cargo, and complete complex tasks using an onboard system. While the industrial use of mobile robots is popular, especially in warehouses and distribution centers, its functions can also be applied to personal assistance and security.

There are a lot of mobile robots which are used for following specific target or object for different tasks like carrying cargos, loads, or just to observe them. Following robot is one of the rapidly growing sub field of the mobile robot. There are also different designs of following robots but wheel one is most commonly used as it is simple, cheaper and efficient than other. These following robots uses combination of hardware and software to recognize the object and to tracking it. For hardware most of the time its camera compatible with processor of robot and for software it depends up on the user to decide which algorithm to use.

The goal of our system is to develop fast and efficient methodology to recognize, track and follow the user or object until user press stop button or bot get lost. In this report we have integrated analytics, design and electronics in order to create optimal system which meets our goal. The system consists of the 4-wheel bot which consist of Pi-Camera for detecting the object. Raspberry pi is as main processor which will process the image captured by Pi- Camera and start tracking the object while giving the necessary command to motors to keep following the object. The process will be continued until the user pressed stop or bot get lost.

1.2 Problem Statement

Transportation of large or heavy objects in places like warehouses or airports done by human is not efficient as humans cannot move such objects for a long distance. Use of bot that can follow human can be used to transport such heavy objects.

Unmanned delivery vehicles could be lost so such vehicles can be tracked using human following bot. The human following bot can only focus on only one target at once.

1.3 Objectives

The main objectives of our project are listed below:

- To train bot to follow specific target to destination.

1.4 Scope and Applications

Scope and application of this project are as follows:

- To observe certain targets.
- Real time object detection, track of the object and sending control signals to follow object.
- The principles of this bot can be used in different sectors when a human and an autonomous robot collaborate on a common task that requires the robot to follow the human. An example application would be the service robots, which are widely used in industrial application, e.g., in manufacturing, warehousing, and health care.

1.5 Organization of project report

This report starts with an abstract which clearly explains the purpose of the project which is further elaborated in introduction. The Chapter 1 contains introduction, i.e. background theory which explains about the concepts on which our project is based. Different objectives and other scopes of the project have been illustrated here. The literature review in Chapter 2 includes the study of related sections and papers. Related theories are in Chapter 3 which has been provided to describe about the components used for the project. The Chapter 4 is project methodology which defines the way in which our project works. Chapter 5 includes the expected result of this project. Chapter 6 includes epilogue which contains problem encountered, future scope and conclusion. We have listed our sources for project ideas in references section.

CHAPTER 2

LITERATURE REVIEW

Robotic technology has increased appreciably in past couple of years. Such innovations were only a dream for some people a couple of years back. But in this rapid moving world, now there is a need of robot such as “A Human Following Robot” that can interact and co-exist with them. [1] To perform this task accurately, robot needs a mechanism that enables it to visualize the person and act accordingly [2]. The robot must be intelligent enough to follow a person in the crowded areas, vivid environment and in indoors and outdoors places the image processing carried out to get the information about the surroundings visually is a very important thing.

The following points should be carefully noted while doing the processing:

- The luminosity conditions should be very stable and should not fluctuate.
- The ranges should be set properly for the desired environment on which to perform the tracking.
- The target should not be very far from the visual sensor as the distance matters a lot.
- We should avoid the use of such colors around the robot that matches with that of the target. Otherwise, the robot would get confused. [3].
- Typically, human following robots are equipped with several different diverse combination of sensors i.e., light detection and ranging sensor, radio frequency identification module (RFID), laser range finder (LFR), infrared (IR) sensing modules, thermal imaging sensors, camera, wireless transmitter/receiver etc. for recognition and locating the target. All the sensors and modules work in unison to detect and follow the target. [4].
- The capability of a robot to track and follow a moving object can be used for several purposes:
 - To help humans in different fields.
 - Can be used for defense purposes.

So far, a lot of research has been done on the kinds of robot that fall into the category of the “Assisting Robots”. People have used different logics and algorithms to implement their design. All of their primary focus has entirely been on the design of robots that follows the target. Laser sensor is used by Burgard in his tour guide robot for human tracking [5]. LRF was incorporated by D. Schulz to perform the “following”. Using the above-mentioned process, they performed the information linking for the detection [6]. Nicola, Husing used a technique for pointing out the different styles of movement by using LRF. This information was fused with the information obtained by the camera [7]. Depth imaging was used by Songmin Jia to carry out the detection. The model of a person was determined using the depth imaging [8]. The particular style of clothing was used by Mehrez Kristou. He used a multidirectional camera. LRF was also incorporated by him in the design [9]. Research was conducted by Wilhelm with the focus on the color of the particular person’s skin. Information from different sensors was also used by him in the research [10].

For a robot that performs autonomously, the communication between the person and the robot is the most important factor. A significant awareness has been observed regarding the usage of such a technology. This research has a trivial involvement in the development of such robots. A robot that functions fully autonomously should not only complete the jobs that are desired of them but also somehow establish a connection between themselves and the person operating them. A lot of research has been done of these kinds of robot and a lot of work still needs to be done. In order for a robot to communicate and interact with the person, it should also be capable of following that particular person. Keeping this in mind, there should be a capacity in the robot to get information from the surroundings while pursuing the required object. The primary goal of our work was to design and fabricate a robot that not only tracks the target but also moves towards it while doing the tracking. In order to make things simpler, a unique handmade tag was placed on the person that the robot needs to follow. The main hindrance in this kind of work is that the detection of the target is a sensitive thing to carry out. The object has to be unique for the robot to recognize it and carry out the objective. The simple tag removes this problem of uniqueness and makes the task fairly easy. A small camera records the video and the processor processes it to extract the desired information from it. Protecting the robot from collision with the object is another problem that needs to be tackled so in order to do this, a sensor is used. All the processing is carried out by the microprocessor while the control of the motors is carried out by the controller.

CHAPTER 3

BACKGROUND THEORY

3.1 Raspberry pi 3B

The Raspberry Pi is a low cost, credit-card sized computer that plugs into a computer monitor or TV, and uses a standard keyboard and mouse. It is a capable little device that enables people of all ages to explore computing, and to learn how to program in languages like Scratch and Python. It's capable of doing everything you'd expect a desktop computer to do, from browsing the internet and playing high-definition video, to making spreadsheets, word-processing, and playing games. The new Raspberry Pi opens up even more possibilities for IoT and embedded projects. With its built-in wireless connectivity, the new Raspberry Pi is clearly positioned as a low-cost hub for Internet of Things devices, or as the flexible, low-cost basis of new types of connected gadgets. The new bump to 2.5 amps power source means it will be able to power more complex USB devices without the need for a second power cable.



Figure 3.1.1: Raspberry-Pi 3B

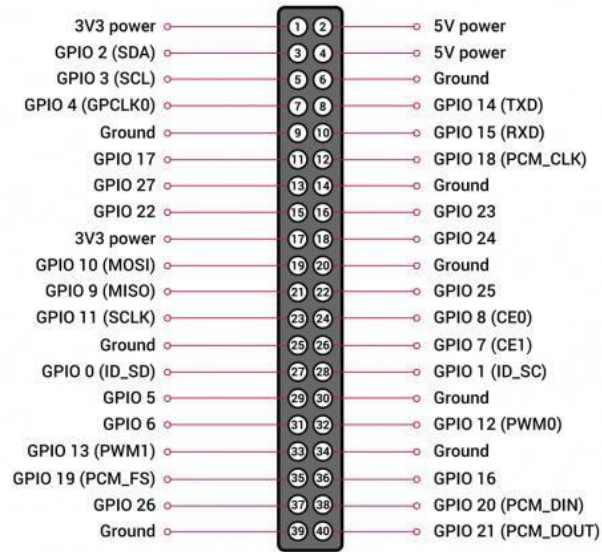


Figure 3.1.2: Pin Configuration of R-pi

Raspberry Pi 3 - Model B Technical Specification:

- Broadcom BCM2387 chipset
- 1.2GHz Quad-Core ARM Cortex-A53
- 802.11 bgn Wireless LAN and Bluetooth 4.1 (Bluetooth Classic and LE)
- 1GB RAM
- 64 Bit CPU
- 4 x USB ports
- 4 pole Stereo output and Composite video port
- Full size HDMI
- 10/100 BaseT Ethernet socket bar
- CSI camera port for connecting the Raspberry Pi camera
- DSI display port for connecting the Raspberry Pi touch screen display
- Micro SD port for loading your operating system and storing data
- Micro USB power source.

3.2 Arduino UNO

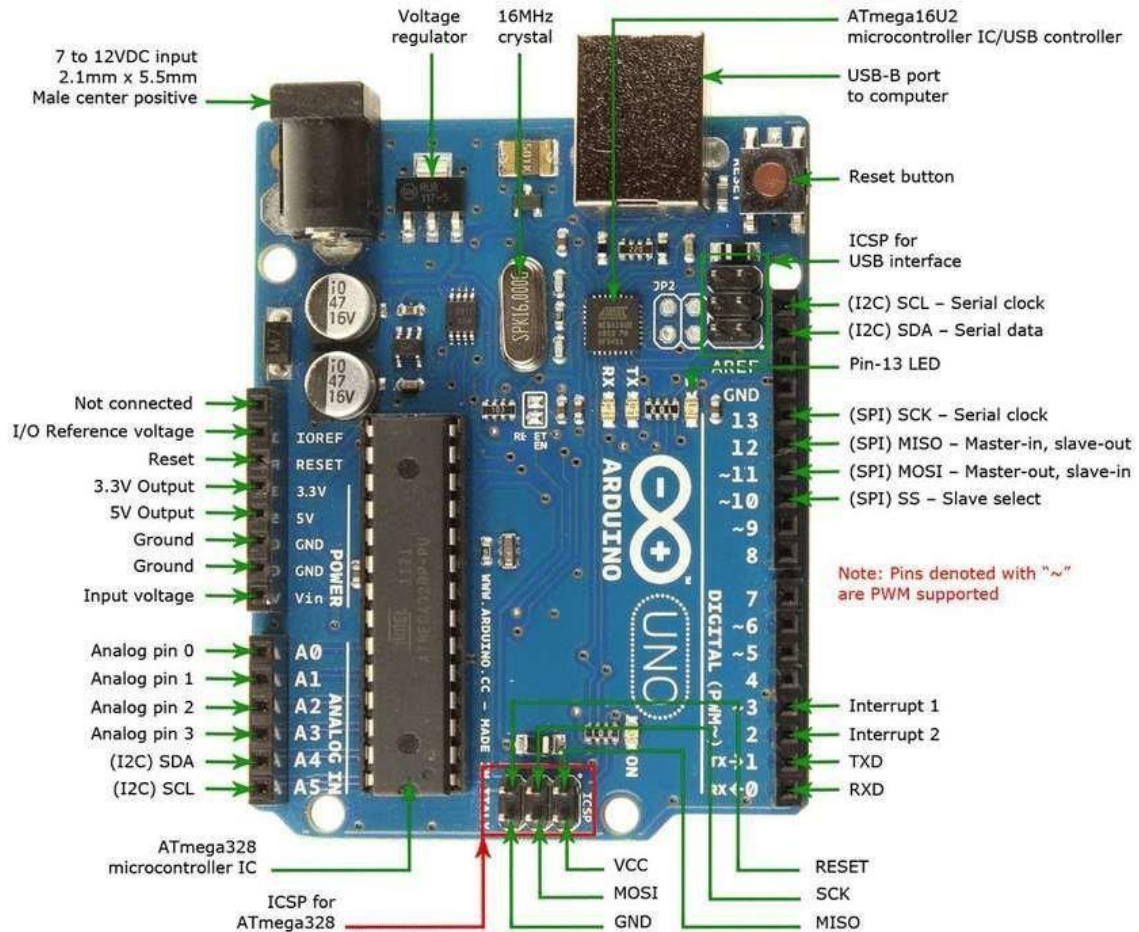


Figure 3.1.3: Pin Configuration of Arduino Uno

Arduino is an open-source, PC paraphernalia and programming organization, Endeavor, and client group that plans and produce microcontroller packs for constructing programmed devices and intelligent object that can detect and control questions in the real world. The inception of the Arduino extends began at the Interaction Design Institute in Ivrea, Italy. The equipment reference plans are appropriated under a Creative Commons Attribution-Share.

Arduino Uno is a microcontroller board based on an 8-bit ATmega328P microcontroller. Along with ATmega328P, it consists of other components such as crystal oscillator, serial communication, voltage regulator, etc. to support the microcontroller. Arduino Uno has 14 digital input/output pins (out of which 6 can be used as PWM outputs), 6 analog input pins, a USB connection, A Power barrel jack, an ICSP header, and a reset button.

3.3 Pi-Camera v2

The Raspberry Pi Camera v2 is the official camera board released by the Raspberry Pi Foundation.



Figure 3.1.4: Pi Camera

Pi-Camera v2 specification:

- Fixed focus lens on-board
- 8-megapixel native resolution sensor-capable of 3280 x 2464-pixel static images
- Supports 1080p30, 720p60 and 640x480p90 video
- Size 25mm x 23mm x 9mm • Weight just over 3g
- Connects to the Raspberry Pi board via a short ribbon cable (supplied)
- Camera v2 is supported in the latest version of Raspbian, Raspberry Pi's preferred operating system.

3.4 Motor Driver(L298N)

This L298N Motor Driver Module is a high-power motor driver module for driving DC and Stepper Motors. This module consists of an L298 motor driver IC and a 78M05 5V regulator. L298N Module can control up to 4 DC motors, or 2 DC motors with directional and speed control.

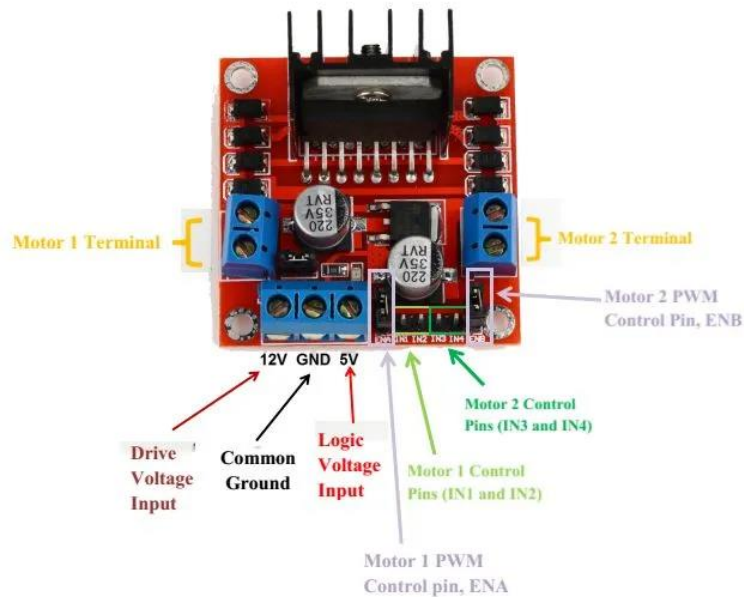


Figure 3.1.5: Motor Driver(L298N)

Features & Specifications:

- Driver Model: L298N 2A
- Driver Chip: Double H Bridge L298N
- Motor Supply Voltage (Maximum): 46V
- Motor Supply Current (Maximum): 2A
- Logic Voltage: 5V
- Driver Voltage: 5-35V
- Driver Current: 2A
- Logical Current: 0-36mA
- Maximum Power (W): 25W
- Current Sense for each motor
- Heatsink for better performance
- Power-On LED indicator

3.5 Power Supply

For the power supply we are using 2200MAH LiPo battery 35C. Which will be sufficient for our project.



Figure 3.1.6: 2200MAH LiPo battery 35C

3.6 DC motor

A DC motor is any of a class of rotary electrical motor that converts direct current electrical energy into mechanical energy. For our project we use Geared Motor 100 rpm 12 V.

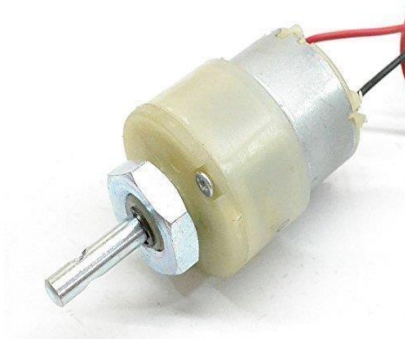


Figure 3.1.7: DC Geared Motor

DC motor specifications:

- Rated Voltage: 12V
- Rated Torque: 6Kg. Cm
- Revolutions Per Minute (RPM): 100rpm
- Working Current: 0.5 A
- Mounting Hole Size:M3
- Outer Diameter: 37mm
- Reduction Ratio: 1:51
- Connection Terminals: 2
- Shaft Diameter: 4 mm

3.7 Buzzer

Figure 3.1.8: Buzzer

A buzzer or beeper is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric.

Buzzer specifications:

- Operating voltage: 3.5-5.5V AC (Alternating Current)
- Rated Voltage: 3 ~ 5 V
- Current Consumption: 25mA
- Resonant Frequency: 2,048 Hz
- Operating Temperature: -20 to 60°C

3.8 Buck converter (LM2596)

A buck converter is a DC-to-DC power converter which steps down voltage from its input to its output. It is a class of switched mode power supply typically containing at least two semiconductors and at least one energy storage element, a capacitor, inductor, or the two in combination.

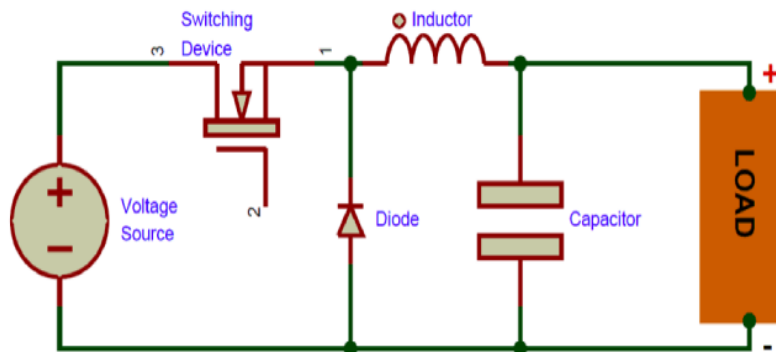
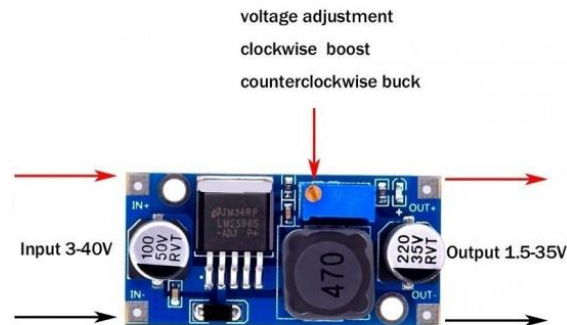


Figure 3.1.9: Buck Converter

Buck Converter Specification:

- Input voltage: 3-40V
- Output voltage: 1.5-35V(Adjustable)
- Output current: Rated current is 2A, maximum 3A(Additional heatsink is required)
- Switching Frequency: 150KHz
- Operating temperature: Industrial grade (-40 to +85)

3.8 Software and programming language descriptions

3.8.1 Python

Python is a high-level, interpreted and general-purpose dynamic programming language that focuses on code readability. The syntax in Python helps the programmers to do coding in fewer steps as compared to Java or C++. ... The Python is widely used in bigger organizations because of its multiple programming paradigms. Created by Guido van Rossum and first released in 1991, Python's design philosophy emphasizes code readability with its notable use of significant whitespace. Its language constructs and object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects. Python is dynamically typed, and garbage collected. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming. Python is often described as a "batteries included" language due to its comprehensive standard library. Python was conceived in the late 1980s as a successor to the ABC language. Image result for why we use python over other languages. There are many advantages of using python over other languages. Which are as follows:

- Presence of Third-Party Modules
- Extensive Support Libraries
- Open Source and Community Development
- Learning Ease and Support Available
- User-friendly Data Structures
- Productivity and Speed

3.8.2 OpenCV

OpenCV (Open-Source Computer Vision Library) is a library of programming functions mainly aimed at real-time computer vision. Originally developed by Intel, it was later supported by Willow Garage then Itseez (which was later acquired by Intel). The library is cross-platform and free for use under the open-source Apache 2 License. Starting with 2011, OpenCV features GPU acceleration for real-time operations.

OpenCV is a great tool for image processing and performing computer vision tasks. It is an opensource library that can be used to perform tasks like face detection, objection tracking, landmark detection, and much more. It supports multiple languages including python, java C++.

3.8.3 Object Tracking with OpenCV

OpenCV offers a number of pre-built algorithms developed explicitly for the purpose of object tracking. There are 8 different trackers available in OpenCV 4.2 — BOOSTING, MIL, KCF, TLD, MEDIANFLOW, GOTURN, MOSSE, and CSRT.

BOOSTING Tracker:

BOOSTING tracker is based on the AdaBoost algorithm of machine learning. The classifier is to be trained at runtime learning on the positive and negative examples of the object to be tracked. It is over a decade old. It is slow and doesn't work very well, even towards some relatively more superficial data.

MIL Tracker:

It is similar in concept to the BOOSTING tracker, with the only difference that instead of only using the current location of the object as a positive example for the classifier, it also looks into a small portion of the neighborhood of the thing. MIL tracker has better accuracy than BOOSTING, but it does a poor job of reporting failure.

KCF Tracker:

It stands for **Kernelized Correlation Filters**. KCF builds on the concept that multiple positive examples in a single bag of MIL tracker have large overlapping regions. The overlap gives rise to some intuitive mathematical approaches for the KCF tracker.

TLD Tracker:

It stands for **Tracking, Learning, and Detection**. This tracker follows the object frame by frame and localizes its position learned from the previous tracking, simultaneously correcting the tracker if necessary.

GOTURN Tracker:

This is the only tracker based on a deep learning approach. It is developed using convolutional neural networks. It is accurate in that it is robust to deformations, lighting changes, and viewpoint changes; at the same time, the downside is that it cannot handle occlusion well.

MOSSE Tracker:

It stands for **Minimum Output Sum of Squared Error**. It used an adaptive correlation for tracking purposes which outputs stable correlation filters. MOSSE was developed between 2010 and 2016 and is robust to scale, pose, non-rigid deformations, and lighting changes. It can also handle occlusion and can instantly resume the tracking when the object reappears.

CSRT Tracker:

CSRT, otherwise known as Discriminative Correlation Filter with Channel and Spatial Reliability (DCF-CSR), used a spatial reliability map to adjust the filter to the part of

the selected frame for tracking. This helps in the localization of the object of interest. It also gives high accuracy for comparatively lower fps (25 fps).

MEDIANFLOW Tracker:

Internally, this tracker tracks the object in both forward and backward directions in time and measures the discrepancies between these two trajectories. Minimizing this Forward Backward error enables them to reliably detect tracking failures and select reliable trajectories in video sequences.

This tracker works best when the motion is predictable and small. Unlike, other trackers that keep going even when the tracking has clearly failed, this tracker knows when the tracking has failed.

Pros: Excellent tracking failure reporting. Works very well when the motion is predictable and there is no occlusion.

Cons: Fails under large motion

3.8.4 Proteus

The Proteus Design Suite is a proprietary software tool suite used primarily for electronic design automation. The software is used mainly by electronic design engineers and technicians to create schematics and electronic prints for manufacturing printed circuit boards.

It was developed in Yorkshire, England by Lab Center Electronics Ltd and is available in English, French, Spanish and Chinese languages.

The first version of what is now the Proteus Design Suite was called PC-B and was written by the company chairman, John Jameson, for DOS in 1988. Schematic Capture support followed in 1990, with a port to the Windows environment shortly thereafter. Mixed mode SPICE Simulation was first integrated into Proteus in 1996 and microcontroller simulation then arrived in Proteus in 1998. Shape based auto routing was added in 2002 and 2006 saw another major product update with 3D Board Visualization. More recently, a dedicated IDE for simulation was added in 2011 and MCAD import/export was included in 2015. Support for high-speed design was added in 2017. Feature led product releases are typically biannual, while maintenance-based service packs are released as it is required.

3.8.5 Serial Communication

Serial communication is the process of sending data one bit at a time, sequentially, over a communication channel or computer bus. Serial is used for communication between the Arduino board and a computer or other devices. To reduce the number of pins in a package, many ICs use a serial bus to transfer data when speed is not important. Some examples of such low-cost serial buses include RS-232, SPI, I²C, and PCI Express. All Arduino boards have at least one serial port (also known as a UART or USART). Serial communication between Raspberry and Arduino to serial communication using the UART protocol, where UART stands for “Universal Asynchronous Reception and Transmission”. Basically, it’s an asynchronous multi-master protocol based on the Serial communication, which will allow to communicate between the two boards. There isn’t a common clock in asynchronous serial communication; instead, both devices keep time independently, and either send or listen for new bits of data at an agreed-upon rate.

There are two ways to connect Raspberry Pi and Arduino for serial communication:

- Serial via GPIO pins (UART, SPI, I²C)
- Serial via USB (UART)

CHAPTER 4

METHODOLOGY

4.1 Block Diagram

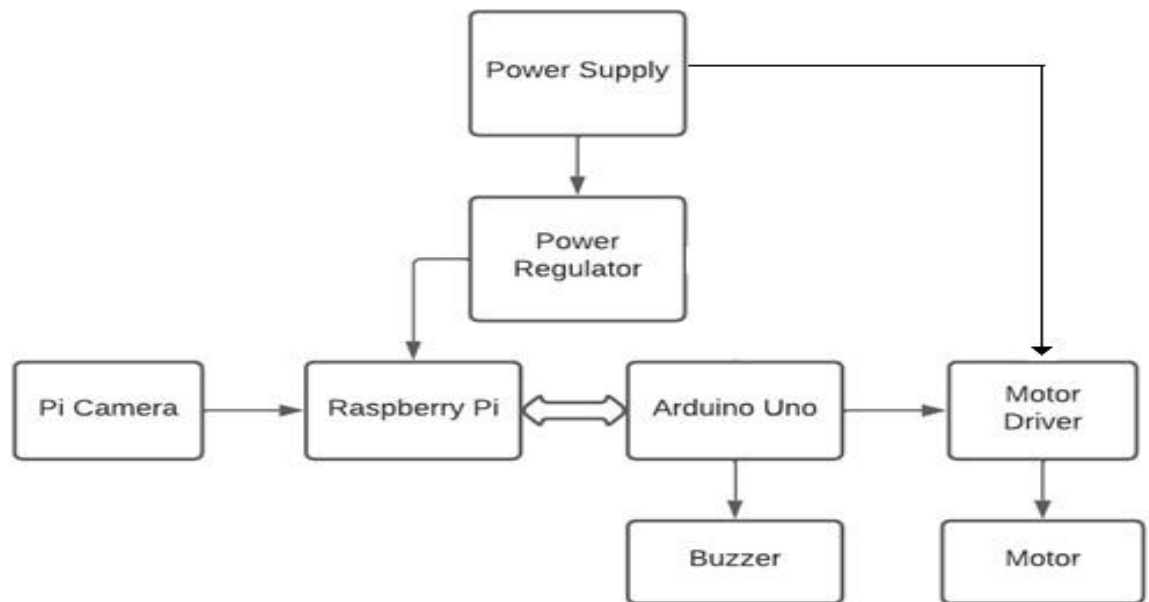


Figure 4.10: Block Diagram of System

Power supply: 12v (2200mah) Li-Po is power supply used as main power supply for our project where regulated 5v power is taken out for supply to Raspberry pi.

Power regulator: We use buck converter (LM2596) is used for voltage regulator for the regulating 12V DC to 5V DC. Which is used as supply to Raspberry pi.

Pi-Camera: Pi-Camera v2 (5mp) is used as main vision for the bot where it captures videos necessary to detect and track the target and fed it to Raspberry pi for further processing.

Raspberry pi: Raspberry pi is used as main processing unit for our project and used to all the detection and tracking of the object. Generate necessary control signal for the control of motors of bot and send necessary signal to Arduino for control of motors.

Arduino Uno: Here Arduino Uno is used as controlling of the all the motors of the bot which can be done according to signal received form Raspberry pi.

Motor driver: Motor driver (L2983D) is used for controlling all the motors according to signal of Arduino while providing them with high current they need.

Buzzer: Here main job of buzzer is to notify user bots location but buzzing when it get lost.

Motor: 12V DC motors are for providing necessary movement of the bot.

4.2 Algorithm

Step1: Start

Step2: Initialize ports as Input

Step3: Read data from Pi-Camera

Step4: Identify the target and start tracking

Step5: Start follow the target and keep following

Step6: If user entered Stop button go to Step 13 else continue

Step7: If target still in sight go to Step 5 else continue

Step8: Start searching for target

Step9: If target found go to Step 5 else continue

Step10: Initialize ports as Output

Step11: Notify the user and wait until user press stop button

Step12: If user press Buzzer button turn on buzzer

Step13: Stop

4.3 Flowchart

4.3.1 System Flowchart:

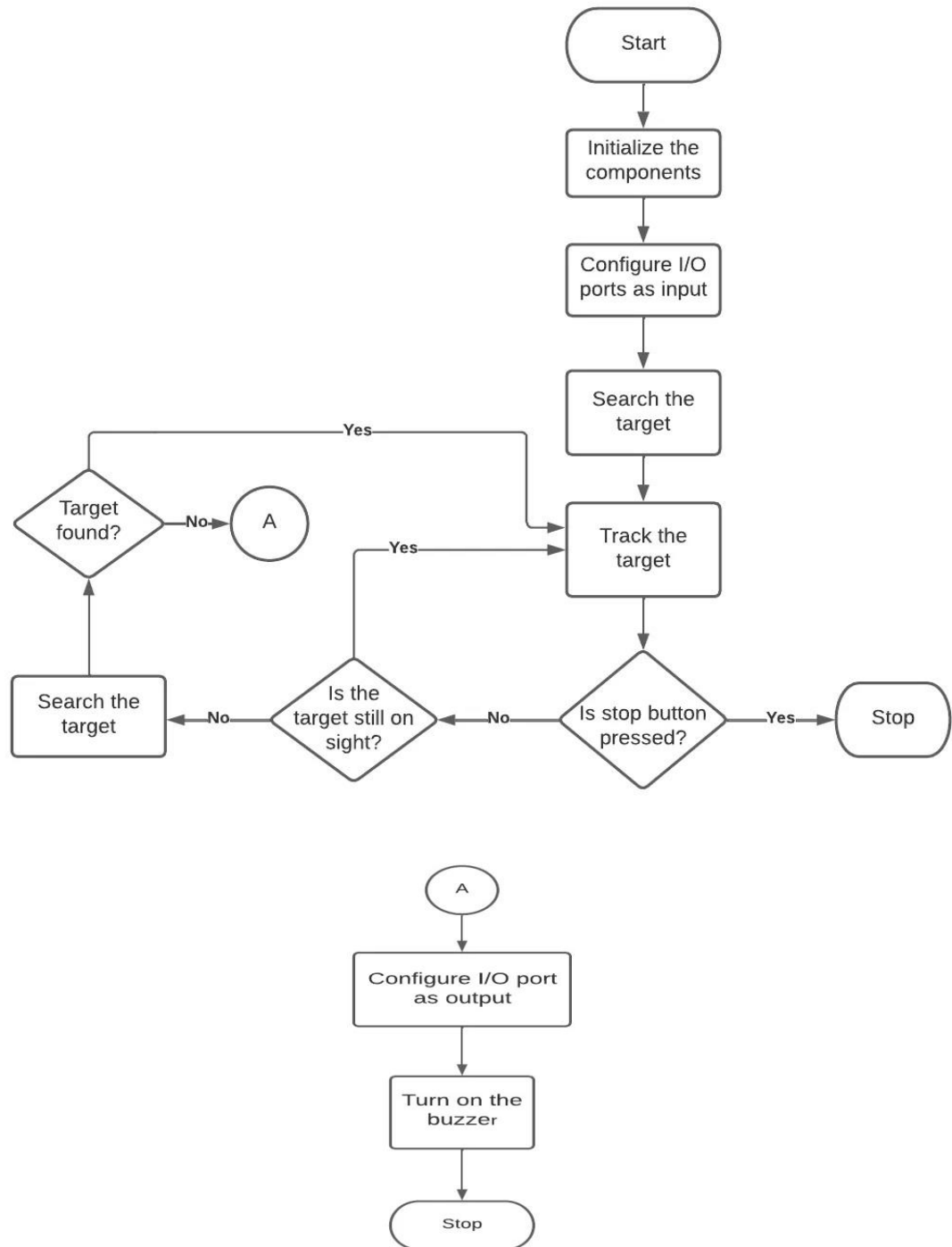
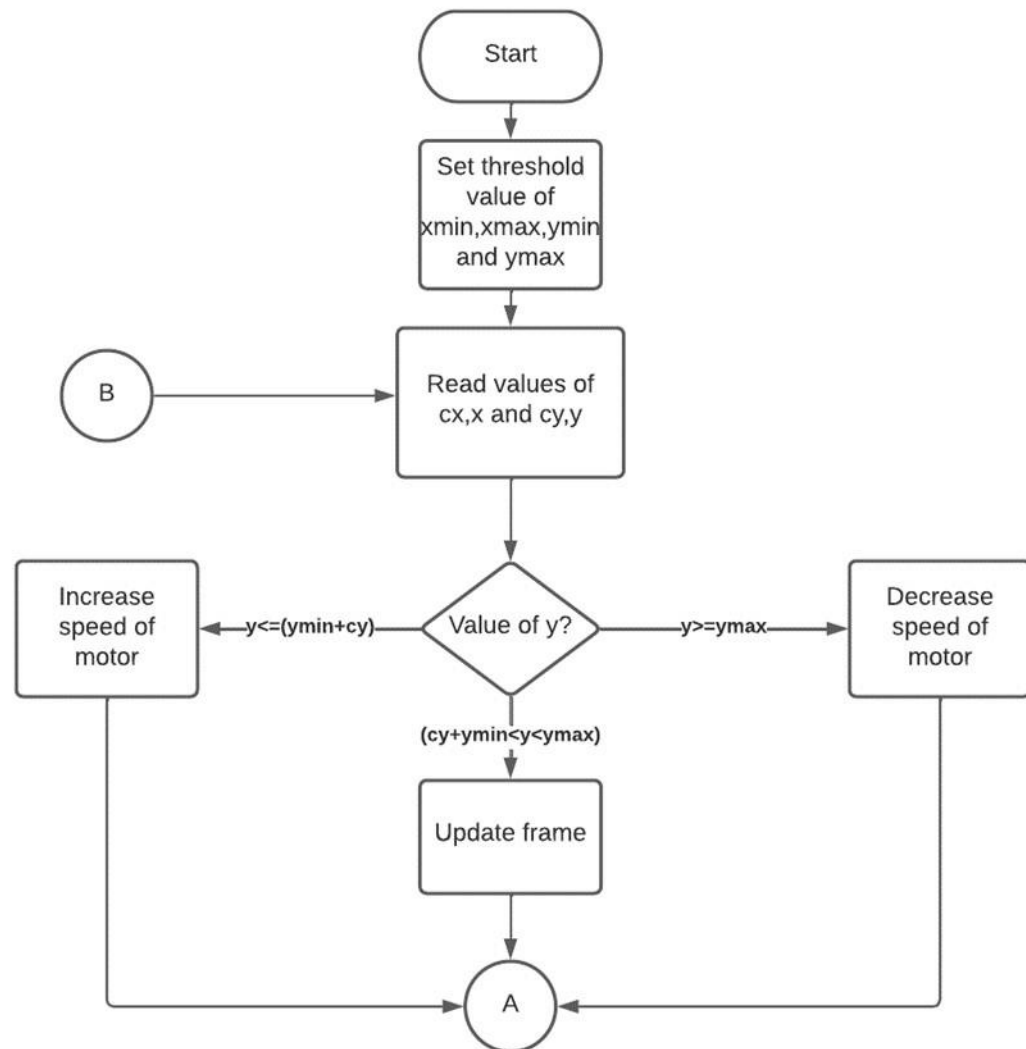


Figure 4.2.11: System Flowchart

4.3.2 Motor Flowchart:



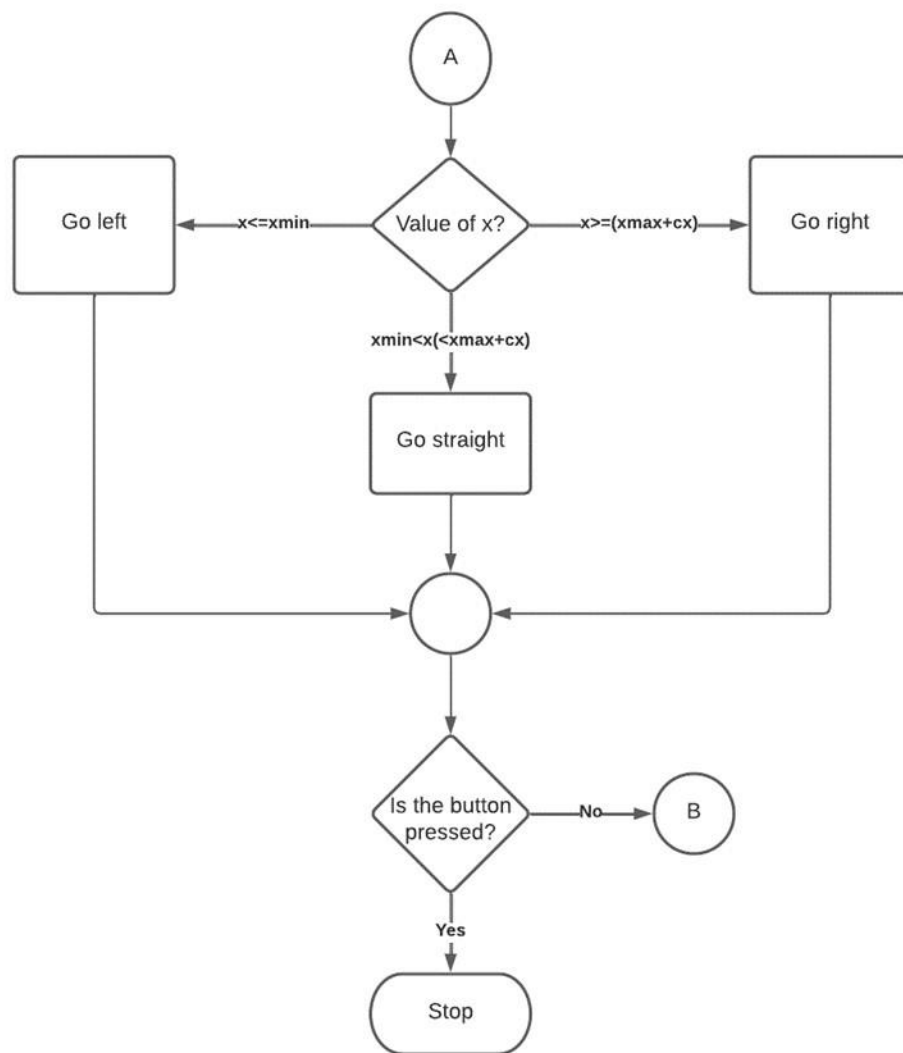
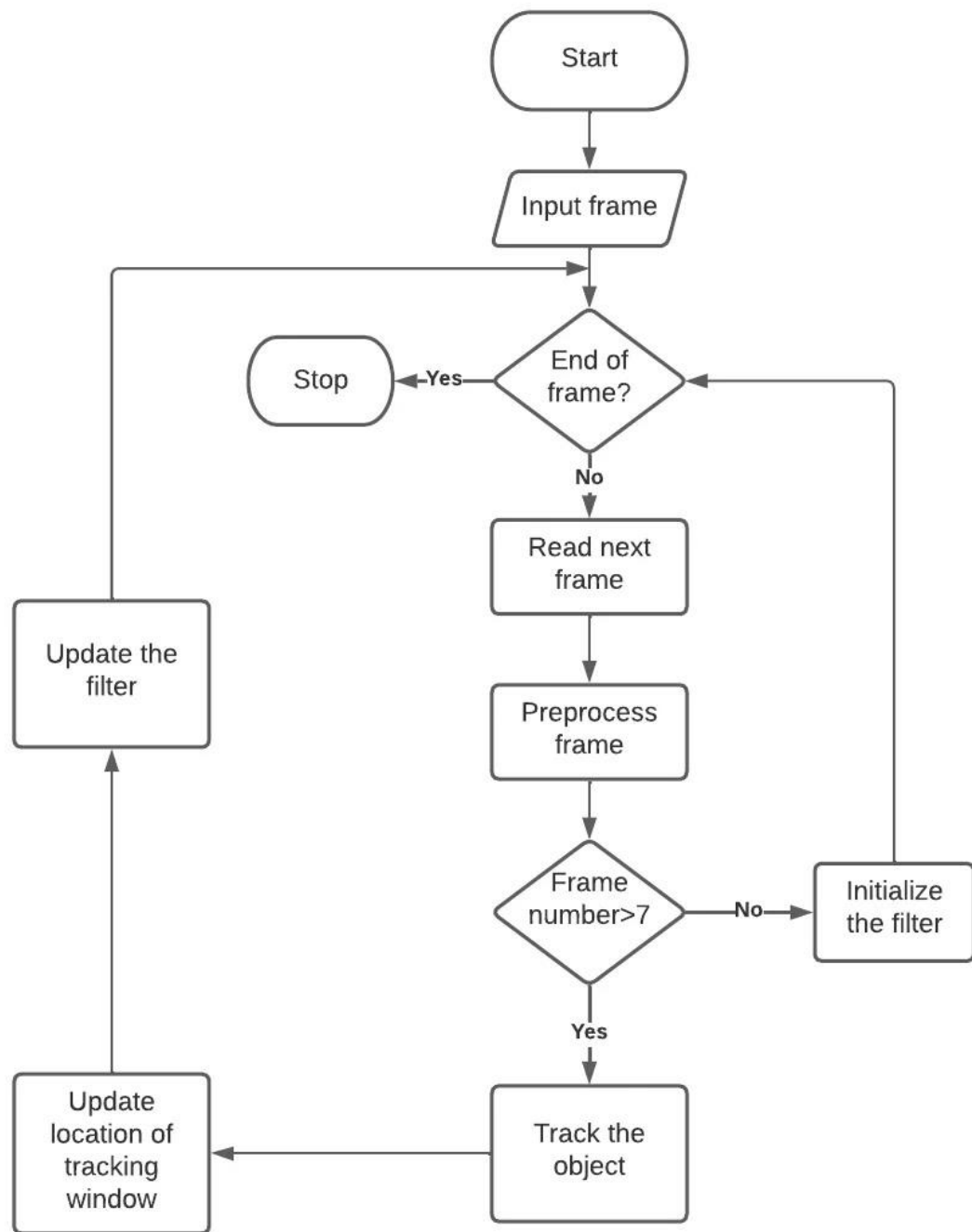


Figure 4.2.12: Motor Flowchart

4.3.3 Tracker Flowchart:



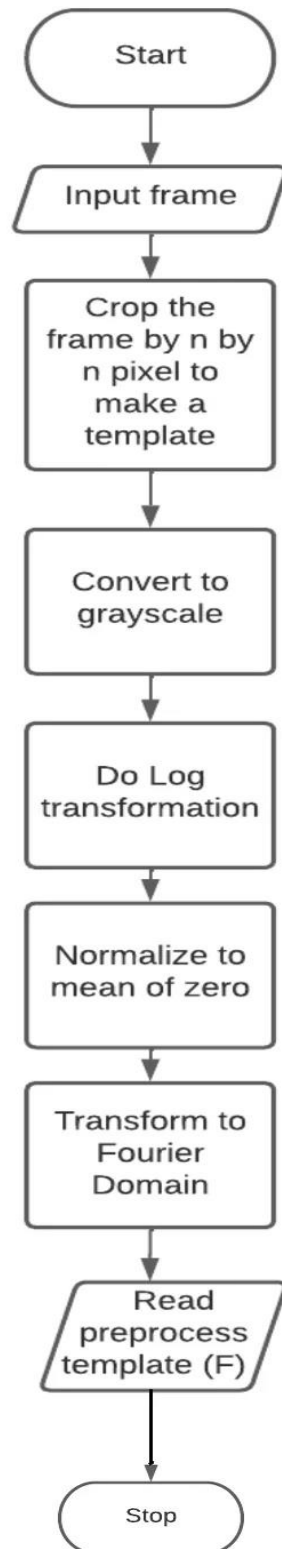


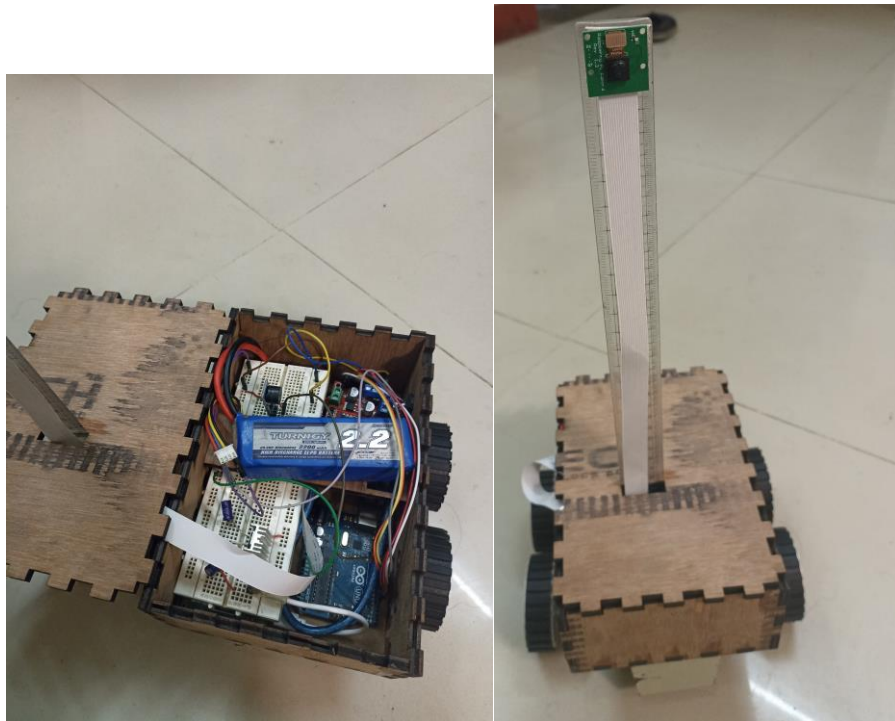
Figure 4.2.13: Tracker Flowchart

CHAPTER 5

RESULT AND ANALYSIS

5.1 Completed Results

We successfully detect and tracked the targeted object using raspberry pi. We did serial communication between Raspberry pi and Arduino. We successfully moved the bot to follow the person according to their movement. The goal of our system is to develop fast and efficient methodology to recognize, track and follow the user or object until user press stop button or bot get lost. In this report we have integrated analytics, design and electronics in order to create optimal system which meets our goal. The system consists of the 4-wheel bot which consist of Pi-Camera for detecting the object. Arduino is used to do control all the sensors and actuators in our project after it receive the data from raspberry pi.



CHAPTER 6

EPILOGUE

6.1 Problem Encountered

Voltage regulator

Problem:

First, we used LM7805 as voltage regulator with following specification.

- Input voltage: 7-35 DC
- Output Voltage: 4.8-5.2 V
- Output current: 1A

But as we use it, we noticed that its limited current (1A) is not enough to operate Raspberry pi (which is serially connected to Arduino uno) at its maximum load.

Solution:

So, we finalized on using buck converter (LM2596) with following specification.

- Input voltage: 4.5-35 DC
- Output Voltage: 1.3-30 V
- Output current: 3A Maximum

Which will provide enough current for Raspberry pi at its maximum load.

Computational power

Problem:

So, at start of project we were finalized to use only Raspberry pi for all processing and controlling of the project. But as we start doing project, we found that Raspberry pi will not be enough for all of these processing and controlling. As it's processing power is limited.

Solution:

So, what we decide to do was use Arduino uno for controlling part and Raspberry pi for processing part. And we will do serial communication between them for successfully execution of our project. That will solve the computational power problem.

Open-CV Build in tracker

Problem:

At starting we discussed about using KCF (Kernelized Correlation Filters) or CSRT (Channel and Spatial Reliability Tracking) tracker. As they are the one of most efficient algorithms which are built in Open-CV. When we run those algorithms on laptop, they run smoothly without any problems. But as we try to run them on Raspberry pi, we encountered that FPS of result is not useable as they give 1-4 fps. It's because those algorithms are very efficient because they use more computational powers for more efficient tracking and as we know computational power of Raspberry pi is very limited. So, we have to change the tracker to solve this problem.

Solution:

As we knew Computational power of Raspberry pi cannot handle those heavy build in trackers so what we need is built in tracker which is not only light enough to give useable FPS on Raspberry pi but also should be efficient enough to be useful in our project. So, at last we finalized on MOSSE (Minimum Output Sum of Squared Error) which is light enough to give useful FPS on Raspberry pi also efficient enough to use on our project.

6.2 Future Scope

Following are the future scope of our project are as following

- They can be used in hospital for following doctors or nurse while carrying medicines.
- They can be used as surveillance bots where it keeps track of object with updating its location to monitoring room.
- Advance version of these can be used as guide for blind people by giving them proper explanation of surrounding.

6.3 Discussion and Conclusion

Firstly, follow me bot is bot which will detect and track the targeted object till it reaches destination and in case if bot can't find targeted person, then it will stop and notify the user.

We are using Raspberry pi and Arduino Uno as our main process and controller. We are using Arduino along with Raspberry pi as raspberry Pi cannot handle all the operations that we did in this project, we have divided the work among Raspberry Pi and Arduino Uno to give the result we wanted.

Raspberry pi is used to process the image taken from raspberry pi camera. For that we have to choose best tracker after doing research optimal algorithm was chosen. The algorithm that got chosen was MOOSE (Minimum Output Sum of Squared Error) algorithm for our project. A bounding box is drawn around the object after selecting an object for labeling. The bounding box represents the template during initialization and the tracking window during tracking and region inside it is objects data. After that another box is created in each frame which is used to know where does targeted object is going and respective data is send to Arduino by raspberry pi through serial port. Arduino is used to do control all the sensors and actuators in our project after it receive the data from raspberry pi.

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